

Solution. See that singularities exist at both ends. Hence, we write

$$I = \int_0^\infty \ln\left(x + \frac{1}{x}\right) \cdot \frac{dx}{1+x^2} = \int_0^1 \ln\left(x + \frac{1}{x}\right) \cdot \frac{dx}{1+x^2} + \int_1^\infty \ln\left(x + \frac{1}{x}\right) \cdot \frac{dx}{1+x^2}$$

provided both integrals at the right be convergent.

Thus, we are only to calculate,

$$\lim_{\epsilon \rightarrow 0^+} \int_\epsilon^1 \ln\left(x + \frac{1}{x}\right) \frac{dx}{1+x^2} + \lim_{B \rightarrow \infty} \int_1^B \ln\left(x + \frac{1}{x}\right) \frac{dx}{1+x^2}$$

[Put $x = \tan \theta$]

$$\begin{aligned} &= - \lim_{\epsilon \rightarrow 0^+} \int_{\tan^{-1} \epsilon}^{\pi/4} (\ln \sin \theta + \ln \cos \theta) d\theta - \lim_{B \rightarrow \infty} \int_{\pi/4}^{\tan^{-1} B} (\ln \sin \theta + \ln \cos \theta) d\theta \\ &= - \int_0^{\pi/4} (\ln \sin \theta + \ln \cos \theta) d\theta - \int_{\pi/4}^{\pi/2} (\ln \sin \theta + \ln \cos \theta) d\theta \end{aligned}$$

[both integrals are convergent]

$$= - \int_0^{\pi/2} (\ln \sin \theta + \ln \cos \theta) d\theta = -2 \cdot \frac{\pi}{2} \cdot \ln \frac{1}{2} = -\pi \ln \frac{1}{2} = \pi \ln 2,$$

by the previous example.

Examples XI(A)

1. Show that:

$$\begin{array}{llll} \text{(a)} \int_0^\infty e^{-x} dx = 1 & \text{(b)} \int_1^\infty \frac{dx}{x^{3/2}} = 2 & \text{(c)} \int_{-\infty}^0 e^{2x} dx = \frac{1}{2} & \text{(d)} \int_0^\infty \frac{dx}{4+9x^2} = \frac{\pi}{12} \\ \text{(e)} \int_{-\infty}^\infty \frac{x dx}{x^4+1} = 0 & \text{(f)} \int_0^3 \frac{dx}{\sqrt{9-x^2}} = \frac{\pi}{2} & \text{(g)} \int_0^\infty \frac{dx}{1+x^2} = \frac{\pi}{2} & \text{(h)} \int_1^\infty \frac{1}{x^2} dx = 1 \\ \text{(i)} \int_{-\infty}^\infty x e^{-x^2} dx = 0 & \text{(j)} \int_0^1 \frac{dx}{\sqrt{1-x^2}} = \frac{\pi}{2}. & & \end{array}$$

[C.H. 1990]

2. Prove that

$$\begin{array}{ll} \text{(a)} \int_0^\infty e^x dx, \int_0^\infty \frac{dx}{x} \text{ and } \int_a^b \frac{dx}{x-a} \text{ (} b > a > 0 \text{) diverge to } \infty \\ \text{(b)} \int_a^\infty \sin x dx, \int_{-\infty}^b \sin x dx \text{ and } \int_{-\infty}^\infty \sin x dx \text{ are oscillatory} \\ \text{(c)} \int_0^c \frac{dx}{x \ln x} \text{ and } \int_c^1 \frac{dx}{x \ln x} \text{ diverge to } -\infty, \text{ for } 0 < c < 1 \\ \text{(d)} \int_0^1 \frac{1}{x} dx; \int_1^\infty \frac{1}{x} dx \text{ and } \int_2^\infty \frac{dx}{x \ln x} \text{ diverge to } \infty \end{array}$$

[C.H. 1990]

(e) $\int_a^\infty e^{-px} dx$ and $\int_{-\infty}^b e^{px} dx$ converge for any constant $p > 0$ and diverge for $p < 0$.

3. (a) Prove that $\int_{-1}^1 \frac{1}{x^3} dx$ exists in Cauchy principal value sense although the integral does not exist.

(b) Prove that Cauchy principal value of $\int_{-1}^1 \frac{1}{x} dx = 0$, $\int_1^2 \frac{dx}{x} = \ln 2$, but of $\int_{-1}^1 \frac{1}{|x|} dx$ does not exist.

(c) Show that $\int_{-1}^2 \frac{dx}{x}$ does not exist, but $P \int_{-1}^2 \frac{dx}{x} = \ln 2$.

$$4. \int_0^\infty \frac{dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{2ab(a+b)}; a, b > 0.$$

$$5. \int_0^\infty \frac{x dx}{(x^2+a^2)(x^2+b^2)} = \frac{1}{a^2-b^2} \ln\left(\frac{a}{b}\right); a, b > 0.$$

$$6. \int_0^\infty \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{2(a+b)}; a, b > 0.$$

$$7. \int_0^\infty e^{-x} \sin x dx = \frac{1}{2}.$$

$$8. \int_0^\infty e^{-ax} \cos bx dx = \frac{a}{a^2+b^2}; a > 0.$$

$$9. \int_0^\infty e^{-ax} \sin bx dx = \frac{b}{a^2+b^2}; a > 0.$$

$$10. \int_0^\pi \frac{dx}{4+5\sin x} = \frac{2}{3} \ln 2.$$

$$11. \int_0^\pi \frac{dx}{a+b\cos x} = \frac{\pi}{\sqrt{a^2-b^2}}, a > b.$$

$$12. \int_0^\pi \frac{dx}{1-2a\cos x+a^2} = \frac{\pi}{1-a^2}, \text{ if } a < 1, \text{ and } = \frac{\pi}{a^2-1}, \text{ if } a > 1.$$

$$13. \int_0^{2\pi} \frac{d\theta}{a+b\cos\theta+c\sin\theta} = \frac{2\pi}{\sqrt{a^2-r^2}}, \text{ if } r^2 = b^2+c^2, r < a.$$

$$14. \int_0^{\pi/2} \frac{dx}{1+\cot x} = \frac{\pi}{4}.$$

$$15. \int_0^\pi \frac{dx}{3+2\sin x+\cos x} = \frac{\pi}{4}.$$

$$16. \int_0^{\pi/2} \frac{dx}{a^2\cos^2 x+b^2\sin^2 x} = \frac{\pi}{2ab}; a, b > 0.$$

$$17. \int_0^{\pi/2} \frac{x \sin x \cos x dx}{(a^2\cos^2 x+b^2\sin^2 x)^2} = \frac{\pi}{4ab^2(a+b)}; a, b > 0.$$

$$18. \int_0^\pi \frac{dx}{(2-\cos x)^2} = \frac{2\pi}{3\sqrt{3}}.$$

[Integrate by parts and use previous example.]

$$19. \int_0^{1/2} \frac{dx}{\sqrt{2x}\{9+(2x)^{1/3}\}} = 3-9\tan^{-1}\frac{1}{3}.$$

$$20. \int_0^1 x^n \ln x dx = -\frac{1}{(n+1)^2}; n > -1.$$

$$21. \int_0^1 \frac{\ln x}{1-x} dx = -\sum_{n=1}^\infty \frac{1}{n^2} = -\frac{\pi^2}{6}.$$

$$22. \int_1^\infty \frac{\ln x}{x^{n+1}} dx = \frac{1}{n^2}.$$

[C.H. 1993]

23. Assuming that the following integrals are convergent prove that

$$(a) \int_0^{\pi/2} \ln \sin x dx = \int_0^{\pi/2} \ln \cos x dx = \frac{\pi}{2} \ln \frac{1}{2}$$

[C.H. 1993]

$$(b) \int_0^\pi \frac{\sin 4x}{\sin x} dx = 0$$

$$(c) \int_0^{\pi/2} \ln \tan x dx = 0$$

$$(d) \int_0^\pi \ln(1+\cos\theta) d\theta = \pi \ln \frac{1}{2}$$

$$(e) \int_0^{\pi} \ln(1 - \cos x) dx = -\pi \ln 2$$

$$(g) \int_0^{\pi/2} \sin 2x \ln \tan x dx = 0$$

$$(i) \int_0^{\pi} x \ln \sin x dx = \frac{\pi^2}{2} \ln \frac{1}{2}$$

$$(k) \int_0^{\pi} \frac{x \tan x dx}{\sec x + \cos x} = \frac{\pi^2}{4}$$

$$(f) \int_0^1 \frac{\ln x}{\sqrt{1-x^2}} dx = \frac{\pi}{2} \ln \frac{1}{2}$$

$$(h) \int_0^{\pi/2} \ln(\tan x + \cot x) dx = \pi \ln 2$$

$$(j) \int_0^{\pi} \frac{x dx}{1 + \cos^2 x} = \frac{\pi^2}{2\sqrt{2}}$$

$$(l) \int_0^{\pi/2} \frac{x dx}{\sec x + \operatorname{cosec} x} = \frac{\pi}{4} \{1 + \ln(\sqrt{2} + 1)\}.$$

24. (a) Show that $\int_0^{\infty} \frac{x}{(1+x)^3} dx = \frac{1}{2} \int_0^{\infty} \frac{dx}{(1+x)^2} = \frac{1}{2}.$

(b) If $a+b > 0$, show that $\int_b^{\infty} \frac{dx}{(x+a)\sqrt{x-b}} = \frac{\pi}{\sqrt{a+b}}.$ [Put $x-b = t^2$.]

(c) Assuming $\int_0^{\pi/2} \ln \sin x dx = \frac{\pi}{2} \ln \frac{1}{2}$, show that $\int_0^{\infty} \ln \left(x + \frac{1}{x}\right) \cdot \frac{dx}{1+x^2} = \pi \ln 2.$

(d) Prove that $\int_1^2 \frac{dx}{(x+1)\sqrt{x^2-1}} = \frac{1}{\sqrt{3}}.$

(e) $\int_{1/2}^1 \frac{dx}{x^4 \sqrt{1-x^2}} = 2\sqrt{3}$

(f) $\int_0^{3a} \frac{2x dx}{(x^2-a^2)^{2/3}} = 9a^{2/3}$

(g) Prove that $\int_0^{\infty} \frac{dx}{x^4 \sqrt{a^2+x^2}} = \frac{2-\sqrt{2}}{3a^4}$ and $\int_0^1 \frac{x^3 \sin^{-1} x}{\sqrt{1-x^2}} dx = \frac{7}{9}.$

25. (a) If $I_n = \int \frac{\sin nx}{\sin x} dx$, show that $(n-1)(I_n - I_{n-2}) = 2 \sin(n-1)x$. Hence or otherwise prove that

$$\int_0^{\pi} \frac{\sin nx}{\sin x} dx = \pi \text{ or } 0 \text{ according as } n \text{ is odd or even.}$$

(b) Show that $\int_0^{\pi} \frac{\sin(n + \frac{1}{2})\theta}{\sin \frac{1}{2}\theta} d\theta = \pi$, if n be a positive integer or zero. What is the value of the integral for negative integral n ?

(c) If $I_n = \int_0^{\pi/2} \frac{\sin nx}{\sin x} dx$, show that $(n-1)(I_n - I_{n-2}) = 2 \sin \frac{1}{2}(n-1)\pi$. Hence, deduce that $I_n = \pi/2$, if n be odd and $= 2 \left\{ 1 - \frac{1}{3} + \frac{1}{5} - \dots + (-1)^{1+n/2} \frac{1}{n-1} \right\}$, if n be even.

(d) If $I_n = \int_0^{\pi} \frac{\sin^2 nx}{\sin^2 x} dx$, when n is a positive integer, show that $I_{n+1} = 2I_n - I_{n-1}$. Hence, deduce that $I_n = n\pi$.

(e) If $I_n = \int_0^{\pi/2} \sin 2nx \cot x \, dx$, $n \geq 1$; show that $I_{n+1} = I_n$ and that $I_n = \frac{\pi}{2}$.

26. Verify each of the following:

$$(a) \lim_{n \rightarrow \infty} \frac{1^a + 2^a + 3^a + \dots + n^a}{n^{a+1}} = \frac{1}{a+1}, a > -1 \quad (b) \lim_{n \rightarrow \infty} \left\{ \frac{1}{n} + \frac{1}{\sqrt{n^2-1^2}} + \dots + \frac{1}{\sqrt{n^2-(n-1)^2}} \right\} = \frac{\pi}{2}$$

$$(c) \lim_{n \rightarrow \infty} \left\{ \frac{1}{\sqrt{2n-1^2}} + \frac{1}{\sqrt{4n-2^2}} + \dots + \frac{1}{n} \right\} = \frac{\pi}{2} \quad (d) \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left\{ 1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} \right\} = 2$$

$$(e) \lim_{n \rightarrow \infty} \sum_{r=1}^{n-1} \frac{1}{n} \sqrt{\left(\frac{n+r}{n-r} \right)} = \frac{\pi}{2} + 1 \quad (f) \lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n} = \frac{1}{e}. \quad [\text{C.H. 1999}]$$

11.2 Necessary and Sufficient Condition for the Convergence of the Improper Integral $\int_a^\infty f(x) \, dx$

Type 1: Absolute convergence.

Let f be bounded and integrable in $a \leq x \leq B$ for every $B > a$ so that the proper integral $\int_a^B f(x) \, dx$ exists and is a function of B , say, $F(B)$. Suppose, now that $F(B)$ tends to a finite limit I as $B \rightarrow \infty$. Then, remembering the definition of a limit, we come to the formal definition:

DEFINITION. Let f be bounded and integrable in $a \leq x \leq B$ for every $B > a$. The integral $\int_a^\infty f(x) \, dx$ is convergent and has the value I , if for any preassigned positive number ϵ , however small, there corresponds a positive number X such that

$$\left| I - \int_a^B f(x) \, dx \right| < \epsilon \text{ for } B > X.$$

Also, restating the *Cauchy criterion* for existence of the limit of the function $F(B)$, we may come to the following theorem.

Theorem 1. Cauchy criterion. A necessary and sufficient condition for the convergence of the integral

$$\int_a^\infty f(x) \, dx$$

is that for any preassigned positive number ϵ , however small, there corresponds a positive number X such that

$$\left| \int_{x'}^{x''} f(x) \, dx \right| < \epsilon$$

for all values of x', x'' for which $x'' > x' > X$.

Clearly, a similar criterion holds for the existence of an improper integral when the singularity is at the lower limit.

Ex. 37. Show that $\int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}}, k^2 < 1$ converges.

Solution. Only singularity is at $x = 1$. By μ -test, it converges since

$$\lim_{x \rightarrow 1-0} (1-x)^{1/2} f(x) = \lim_{x \rightarrow 1-0} (1-x)^{1/2} \cdot \frac{1}{\sqrt{(1-x)(1+x)(1-k^2x^2)}} = \frac{1}{\sqrt{2(1-k^2)}}; \mu = \frac{1}{2} < 1.$$

Ex. 38. Discuss the convergence of $\int_0^1 f dx$, when f is defined on $(0, 1)$ by $f(x) = (-1)^{n+1} \cdot n(n+1), \frac{1}{n+1} \leq x < \frac{1}{n}, n \in \mathbb{N}$.

Solution. When $n = 1, f(x) = 1 \cdot 2$ for $\frac{1}{2} \leq x < 1$

$$n = 2, f(x) = -2 \cdot 3 \text{ for } \frac{1}{3} \leq x < \frac{1}{2}$$

$$n = 3, f(x) = 3 \cdot 4 \text{ for } \frac{1}{4} \leq x < \frac{1}{3}, \text{ etc.}$$

Now,

$$\begin{aligned} \int_{1/(n+1)}^1 f(x) dx &= \int_{1/2}^1 f dx + \int_{1/3}^{1/2} f dx + \int_{1/4}^{1/3} f dx + \cdots + \int_{1/(n+1)}^{1/n} f dx \\ &= \int_{1/2}^1 1 \cdot 2 dx - \int_{1/3}^{1/2} 2 \cdot 3 dx + \int_{1/4}^{1/3} 3 \cdot 4 dx + \cdots + (-1)^{n+1} \int_{1/(n+1)}^{1/n} n(n+1) dx \\ &= \underbrace{1 - 1 + 1 - 1 + \cdots + (-1)^{n+1}}_{\begin{cases} 0, & n \text{ even} \\ 1, & n \text{ odd,} \end{cases}} \end{aligned}$$

which does not exist as $n \rightarrow \infty$. $\therefore \int_0^1 f(x) dx$ does not converge.

Ex. 39. Let f be defined on $0 \leq x < \infty$ by $f(x) = (-1)^{n-1}$, for $n-1 \leq x < n, n \in \mathbb{N}$. Show that $\int_0^\infty f(x) dx$ does not converge.

Solution. As before,

$$\begin{aligned} \int_0^n f(x) dx &= \int_0^1 f dx + \int_1^2 f dx + \int_2^3 f dx + \cdots + \int_{n-1}^n f dx \\ &= \int_0^1 1 dx - \int_1^2 1 dx + \int_2^3 1 dx + \cdots + (-1)^{n-1} \int_{n-1}^n 1 dx \end{aligned}$$

and the result follows by the previous example.

Examples XI(B)

1. Show that the following integrals converge:

(a) $\int_0^1 \frac{dx}{(1+x)\sqrt{x}}$

(b) $\int_0^1 \frac{dx}{(x^2+x)^{1/3}}$

(c) $\int_0^1 \frac{dx}{x^{1/2}(1-x)^{1/3}}$

(d) $\int_0^1 \frac{\ln x}{\sqrt{x}} dx$

(e) $\int_0^\infty \frac{dx}{e^x + 1}$

(f) $\int_0^\infty \frac{\cos x dx}{1+x^2}$

$$\begin{aligned}
 & \text{(g)} \int_0^\infty \frac{\sin x \, dx}{1 + \cos x + e^x} \quad \text{(h)} \int_1^\infty \frac{(x-1)\sqrt{x}}{1+x+x^3+\sin x} \, dx \quad \text{(i)} \int_0^1 \frac{\ln x}{(1-x)^{3/2}} \, dx \\
 & \text{(j)} \int_0^\infty \frac{x^2}{\sqrt{x^7+1}} \, dx \quad \text{(k)} \int_0^\pi \frac{dx}{\sqrt{\sin x}} \quad \text{(l)} \int_1^\infty \frac{\ln x}{x^2} \, dx.
 \end{aligned}$$

2. Show that the following integrals are non-convergent:

$$\begin{aligned}
 & \text{(a)} \int_a^b \frac{dx}{(x-a)\sqrt{b-x}} \quad \text{(b)} \int_2^\infty \frac{dx}{\ln x} \quad \text{(c)} \int_1^2 \frac{\sqrt{x}}{\ln x} \, dx \\
 & \text{(d)} \int_0^\pi \frac{\sin x}{x^3} \, dx \quad \text{(e)} \int_1^\infty \frac{\ln x}{x+a} \, dx, \, a > 0 \quad \text{(f)} \int_0^\pi \frac{dx}{1-\cos x} \\
 & \text{(g)} \int_1^\infty \frac{dx}{x \ln x} \quad \text{(h)} \int_0^\infty \frac{x^{3/2}}{b^2 x^2 + c^2} \, dx.
 \end{aligned}$$

3. (a) Discuss the convergence of the gamma function $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} \, dx$. [C.H. 1996]

(b) Show that the beta function $B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} \, dx$ converges for $m > 0, n > 0$. [C.H. 1999]

(c) Show that $\int_0^{\pi/2} \cos^l x \sin^m x \, dx$ is convergent, if and only if $l > -1, m > -1$. [C.H. 1991]

(d) Show that $\int_0^\infty e^{-x^2} \, dx$ converges. [C.H. 1992]

(e) Show that $\int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2 x^2)}}$, $k^2 < 1$ converges. [C.H. 2000]

(f) Prove that $\int_0^\infty \frac{\sin x}{x} \, dx$ converges but $\int_0^\infty \left| \frac{\sin x}{x} \right| \, dx$ diverges.

(g) Show that $\int_0^1 \frac{1}{x} \sin \frac{1}{x} \, dx$ is conditionally convergent.

4. If f be continuous for $x \geq a$ and $\lim_{x \rightarrow \infty} x^\mu f(x) = \lambda$, then prove the following:

(a) if $\mu > 1$, $\int_a^\infty f(x) \, dx$ converges absolutely,

(b) if $\mu = 1$ and $\lambda \neq 0$, $\int_a^\infty f(x) \, dx$ diverges.

Considering $\int_1^\infty \frac{dx}{x^2}$ and $\int_2^\infty \frac{dx}{x \ln x}$, state what will be your conclusion in case (ii) if $\lambda = 0$. [C.H. 1998]

5. Show that $\int_0^1 \frac{x^{m-1}}{1+x} \, dx$ converges for $m > 0$ and $\int_1^\infty \frac{x^{m-1}}{1+x} \, dx$ converges for $m < 1$ and hence [C.H. 1991]

$\int_0^\infty \frac{x^{m-1}}{1+x} \, dx$ converges for $0 < m < 1$.

6. Prove that $\int_0^\infty \frac{\cos x}{\sqrt{1+x^3}} \, dx$ converges absolutely while $\int_0^\infty \frac{dx}{\sqrt{1+2x^2}}$ diverges.

7. Show that $\int_0^\infty \frac{1 - \cos x}{x^2} dx$ and $\int_0^{\pi/2} \frac{dx}{(\cos x)^{1/n}}, n > 1$ converge.

8. (a) Show that $\int_0^{\pi/2} \frac{\sin^m x}{x^n} dx$ converges, if $n < (m+1)$.

(b) Show that $\int_1^\infty \frac{\sin x}{x^p} dx$ converges absolutely for $p > 1$, but only conditionally if $0 < p \leq 1$.
[C.H. 1997; B.H. 2001]

9. (a) Discuss the convergence of $\int_0^1 x^{n-1} \ln x dx$. What restriction will you impose on n ?

(b) Show that the improper integral $\int_a^\infty \frac{1}{x^\mu} dx (a > 0)$ exists, if $\mu > 1$ and does not exist, if $\mu \leq 1$.
[C.H. 1999]

Use the above result to test the convergence of $\int_{e^2}^\infty \frac{dx}{x \ln \ln x}$.
[C.H. 1995]

(c) Show that $\int_a^b \frac{dx}{(x-a)^\mu}$ converges, if and only if $\mu < 1$.

Use the above result to prove that $\int_0^1 \frac{\ln x}{\sqrt{x}} dx$ converges.

10. (a) If ϕ be bounded when $x \geq a > 0$ and $p > 1$, show that $\int_a^\infty \frac{|\phi(x)|}{x^p} dx$ converges.

(b) Show that $\int_0^\infty \frac{\sin^2 x}{x^2} dx$ converges and is equal to $\int_0^\infty \frac{\sin x}{x} dx$. [Integrate by parts]

(c) Show that $\int_0^\infty \frac{\cos mx}{a^2 + x^2} dx$ converges and $\left| \int_0^\infty \frac{\cos mx}{a^2 + x^2} dx \right| \leq \frac{\pi}{2a}$.
[C.H. 1991]

11. (a) Using the substitution $x = e^{-u}$, show that $\int_0^1 x^{m-1} (\ln x)^n dx$ converges, provided that $m > 0$ and $n > -1$.

(b) By means of a similar substitution as in (i), show that $\int_1^\infty x^{m-1} (\ln x)^n dx$ converges, if $m < 0$ and $n > -1$.

12. Show that $\int_2^\infty \frac{\cos x}{(x + \sin^2 x)(\ln x)^2} dx$ converges, whereas $\int_2^\infty \frac{dx}{(x + \sin^2 x) \ln x}$ diverges.

13. On integrating $\int_1^x \cos x \ln x dx$ by parts, we obtain $\int_1^x \cos x \ln x dx = \sin x \ln x - \int_1^x \frac{\sin x}{x} dx$.

Deduce that $\int_1^x \cos x \ln x dx$ oscillates infinitely. Also, show that $\int_0^1 \cos x \ln x dx$ converges and is equal to $-\int_0^1 \frac{\sin x}{x} dx$.

14. On integrating $\int_{x'}^{x''} \cos x^2 dx$ by parts, we obtain

$$\int_{x'}^{x''} \cos x^2 dx = \frac{1}{2x''} \sin x''^2 - \frac{1}{2x'} \sin x'^2 + \frac{1}{2} \int_{x'}^{x''} \frac{\sin x^2}{x^2} dx,$$

where $x'' > x' > 0$. Deduce the convergence of $\int_0^\infty \cos x^2 dx$.

15. Show that $\int_0^\infty \sin x^2 dx$ and $\int_0^\infty \cos x^2 dx$ converge. [C.H. 2001]

16. Verify that $\int_0^\infty \frac{x dx}{1+x^6 \sin^2 x}$ converges and $\int_0^\infty \frac{x dx}{1+x^4 \sin^2 x}$ diverges.

17. Show that $\int_0^\infty \frac{dx}{1+x^4 \sin^2 x}$ converges but $\int_0^\infty \frac{dx}{1+x^2 \sin^2 x}$ does not.

18. Does $\int_0^{\pi/2} \ln \sin x dx$ converge? — if so, find its value. [C.H. 1993]

19. Assuming $\int_0^{\pi/2} \ln \sin x dx = \frac{\pi}{2} \ln \frac{1}{2}$, show that

(a) $\int_0^\pi \ln(1 - \cos x) dx$ converges to $\pi \ln \frac{1}{2}$ (b) $\int_0^\infty \ln \left(x + \frac{1}{x} \right) \cdot \frac{dx}{1+x^2} = \pi \ln 2$.

20. (a) Show that $\int_1^\infty \frac{\sqrt{x}}{(1+x)^2} dx$ converges to $\frac{1}{2} + \frac{\pi}{4}$. [C.H. 1991]

(b) Prove that $\int_0^1 \frac{dx}{(1+x)(2+x)\sqrt{x(1-x)}}$ converges to $\pi \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{6}} \right)$. [Put $x = \sin^2 \theta$]

[C.H. 2001]

(c) Apply second mean-value theorem for integrals to prove that $\int_x^{x'} \frac{\sin x}{x} dx \rightarrow 0$ as $X', X \rightarrow \infty$.

(d) Show that $\int_0^\infty \frac{dx}{(x + \sqrt{1+x^2})^n} = \frac{n}{n^2-1}$ ($n > 1$).

21. Prove that $\int_0^a \frac{1}{\sqrt{x}} \sin \left(\frac{1}{x} \right) dx$ converges for $0 < a < \infty$.

22. Use Dirichlet's test to prove that $\int_0^\infty \frac{\sin x}{x} dx$ converges. Then use Abel's test to prove that

$$\int_0^\infty e^{-ax} \frac{\sin x}{x} dx, a \geq 0 \text{ is convergent.}$$

23. Apply Dirichlet's test to establish the convergence of

(a) $\int_0^\infty \frac{1}{x} \sin ax \cos bx dx; a > 0, b > 0$ [C.H. 1990] (b) $\int_1^\infty \frac{\sin x}{x^p} dx; p > 0$ (c) $\int_e^\infty \frac{\sin x \ln x}{x} dx$

(d) $\int_0^1 \frac{1}{x^{3/2}} \sin \left(\frac{1}{x} \right) dx$ [C.H. 1992] (e) $\int_0^\infty \sin(x^2) dx$ (f) $\int_a^\infty \frac{\cos x}{\ln x} dx, a > 0$.

24. Show that the following integrals are absolutely convergent: $\int_0^\infty e^{-a^2 x^2} \cos bx \, dx$ and $\int_0^\infty e^{-a^2 x^2} x^m \sin nx \, dx$ ($m > 0$).
25. Show that if $n \geq 3$, $I_n = \int_0^\infty \frac{dx}{\cosh^n x}$ is convergent and is equal to $\frac{n-2}{n-1} I_{n-2}$ and deduce that $I_5 = \frac{3\pi}{16}$.
26. Show that $I_{m,n} = \int_0^\infty \frac{x^m dx}{(1+x)^{m+n}}$ converges when m and $n-1$ are positive integers. Also, prove that $(m+n-1)I_{m,n} = mI_{m-1,n}$. Deduce that $I_{m,n} = \frac{m!(n-2)!}{(m+n-1)!}$.
27. Evaluate $\int_0^\infty \frac{dx}{(1+x^2)^n}$ and $\int_0^\infty \frac{dx}{(1+x^2)^{n+\frac{1}{2}}}$, n being a positive integer.
28. Show that $\int_0^h \frac{\sin x}{x^p} dx$, where $h > 0$ is convergent, if $p < 2$.
29. (a) Let f be defined on $(0, 1)$ by $f(x) = (-1)^{n+1} \cdot n(n+1)$ for $\frac{1}{n+1} \leq x < \frac{1}{n}$, $n \in \mathbb{N}$.
Show that $\int_0^1 f(x) dx$ does not converge. [C.H. 1996]
- (b) Let f be defined on $0 \leq x < \infty$ by $f(x) = (-1)^{n-1}$, for $n-1 \leq x < n$, $n \in \mathbb{N}$.
Show that $\int_0^\infty f(x) dx$ does not exist.
30. f is defined on $0 \leq x < \infty$ by $f(x) = \begin{cases} 1, & \text{for } n \leq x \leq n + \frac{1}{n^2}; \, n = 1, 2, 3, \dots \\ 0, & \text{otherwise.} \end{cases}$
Show that $\int_0^\infty f(x) dx$ exists although $\lim_{x \rightarrow \infty} f(x)$ does not.
31. Prove that $\int_0^\infty \frac{x^{s-1}}{1+x^t} dx$ is convergent, if and only if $0 < s < t$.
32. Show that $\int_0^\infty \frac{x^{s-1} - x^{t-1}}{1-x} dx$ converges, if and only if $0 < s < 1$ and $0 < t < 1$.
33. Show that $\int_0^{\pi/2} \cos 2nx \ln \sin x \, dx$ converges and is equal to $-\pi/4n$, when n is a positive integer.
34. (a) From the relation $\frac{\sin 2nx}{\sin x} = \sum_{r=1}^n \cos(2r-1)x$, show that $\int_0^{\pi/2} \frac{\sin 2nx}{\sin x} dx = 2 \sum_{r=1}^n \frac{(-1)^{r-1}}{2r-1}$.
Deduce that $\lim_{n \rightarrow \infty} \int_0^{\pi/2} \frac{\sin 2nx}{\sin x} dx = \frac{\pi}{2}$.
- (b) By integration by parts, show that $\lim_{n \rightarrow \infty} \int_0^{\pi/2} \sin 2nx \left\{ \frac{1}{\sin x} - \frac{1}{x} \right\} dx = 0$.
- (c) From the above, prove that $\int_0^{\pi/2} \frac{\sin x}{x} dx = \frac{\pi}{2}$.